

ScamShield AI

Detecting Scams, Protecting People

AI-Powered • Real-time • On-device

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You've WON \$5,000! Claim
now: bit.ly/xyz123



Scanning message...



Likely Scam
94% confidence



Hey, are we still on for lunch?

Project Objectives

- The primary objective of **ScamShield AI** is to develop an AI-powered WhatsApp scam detection system specifically designed for the **Indian digital ecosystem**.
- The proposed system focuses on identifying fraudulent messages written in **English, Hindi, Gujarati, and Hinglish**, addressing a significant gap in existing scam detection solutions, which primarily support English-language content.

What does this service do?

It detects phishing, UPI fraud, fake KYC, digital arrest, lottery, and other common scams targeting Indian users.

How does it work?

It analyzes multilingual messages (English, Hindi, and Hinglish) using AI and Machine Learning, classifying them as Safe, Suspicious, or Scam with a confidence score.

What kind of results does it provide?

It provides explainable results by highlighting suspicious phrases and the reason for detection.

Is it easy to use?

Yes, it enables instant and easy scam verification for users without technical expertise.

Is it up-to-date?

Yes, it builds a scalable platform with a continuously updated database of emerging scam patterns.



Literature Review

Tool	Type	Key Characteristic
Norton Genie	Standalone app (paid)	English-only; explicitly does not evaluate scams in other languages
Bitdefender Scamio	WhatsApp/web chatbot (free)	Requires account signup mid-flow; does not analyze non-English text, explicitly asking users to translate messages into English first
Trend Micro ScamCheck	Web tool (limited free)	Character limit issues; missed phishing scams in independent tests
McAfee Scam Detector	Bundled in security suite	Full features gated behind subscription; strong on links, weaker on plain text
Google Messages / Phone AI	On-device (Android)	Built-in, proactive, but general-purpose not India-scam-specialized
Meta AI (in WhatsApp)	General assistant	Reactive user must think to ask; not a dedicated scam-checking flow

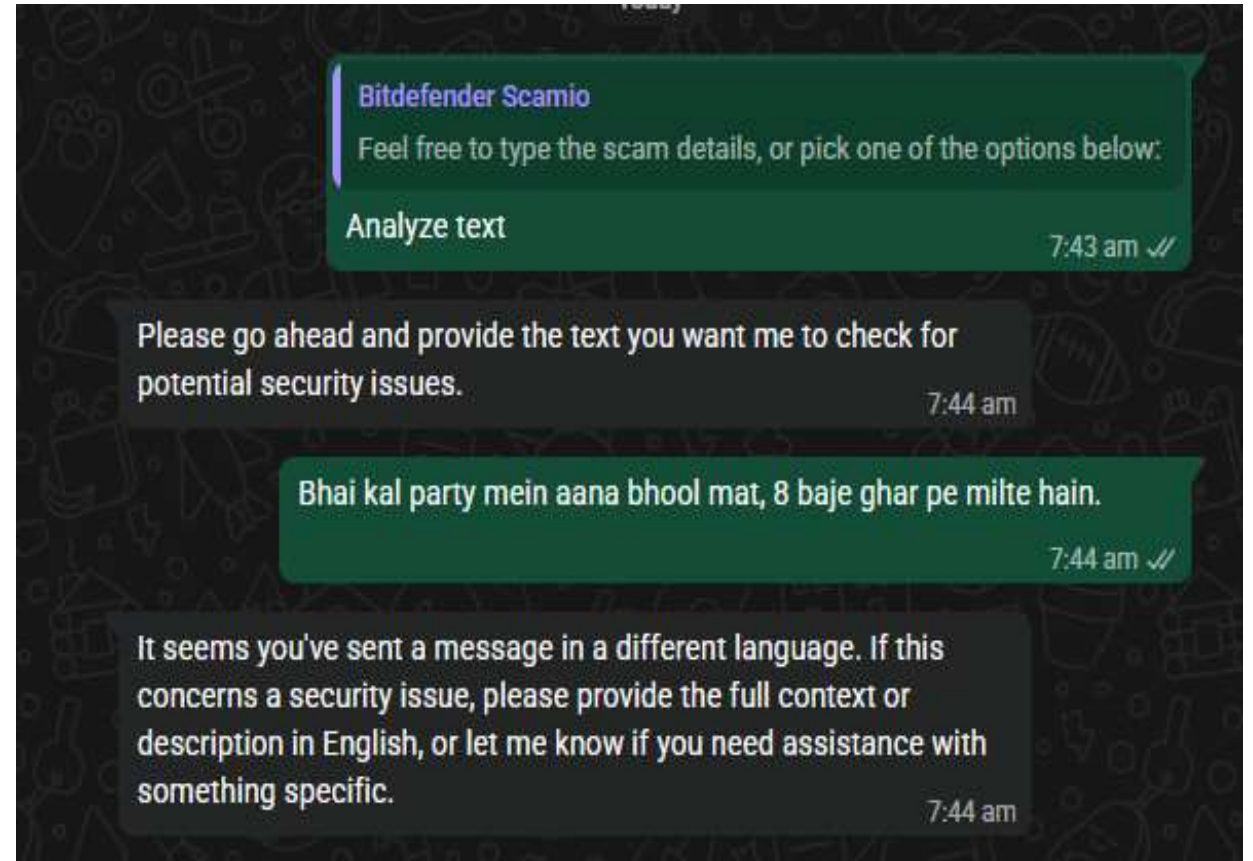
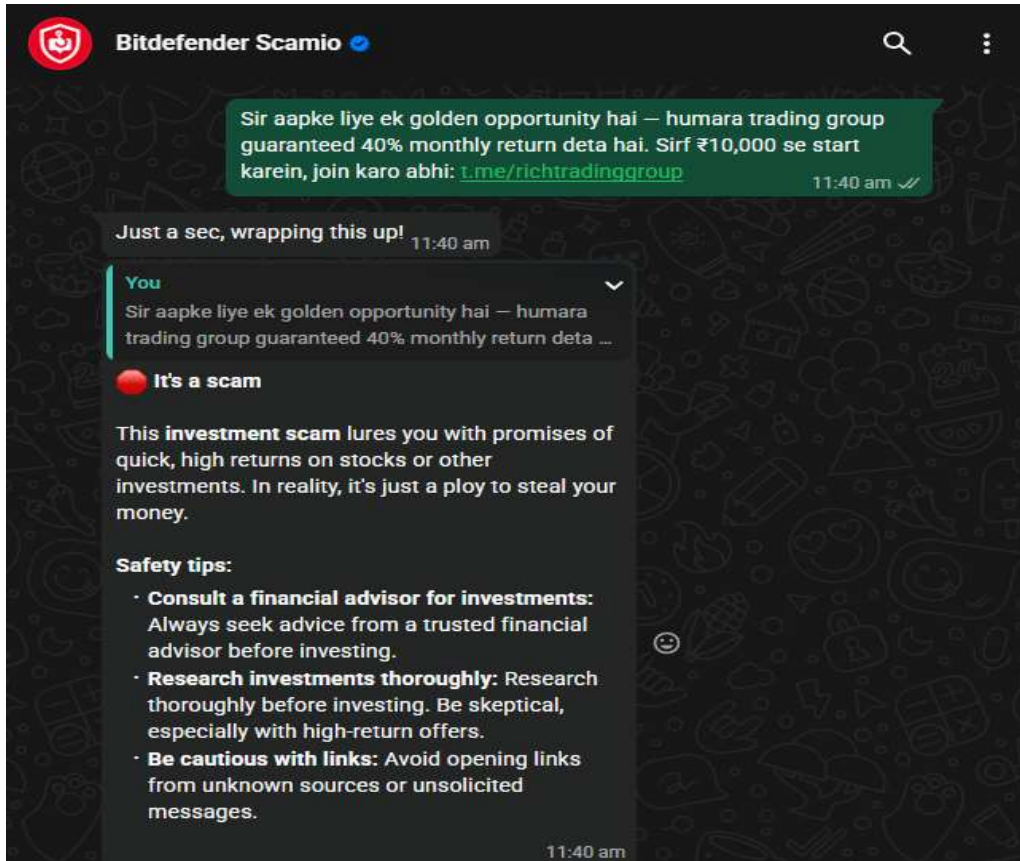
Sources: official product pages, independent reviewer testing (Safety Detectives, KSL News), and hands-on testing conducted for this project.

Research Gap

We didn't just read reviews we tested Bitdefender Scamio directly with real scam-style messages.

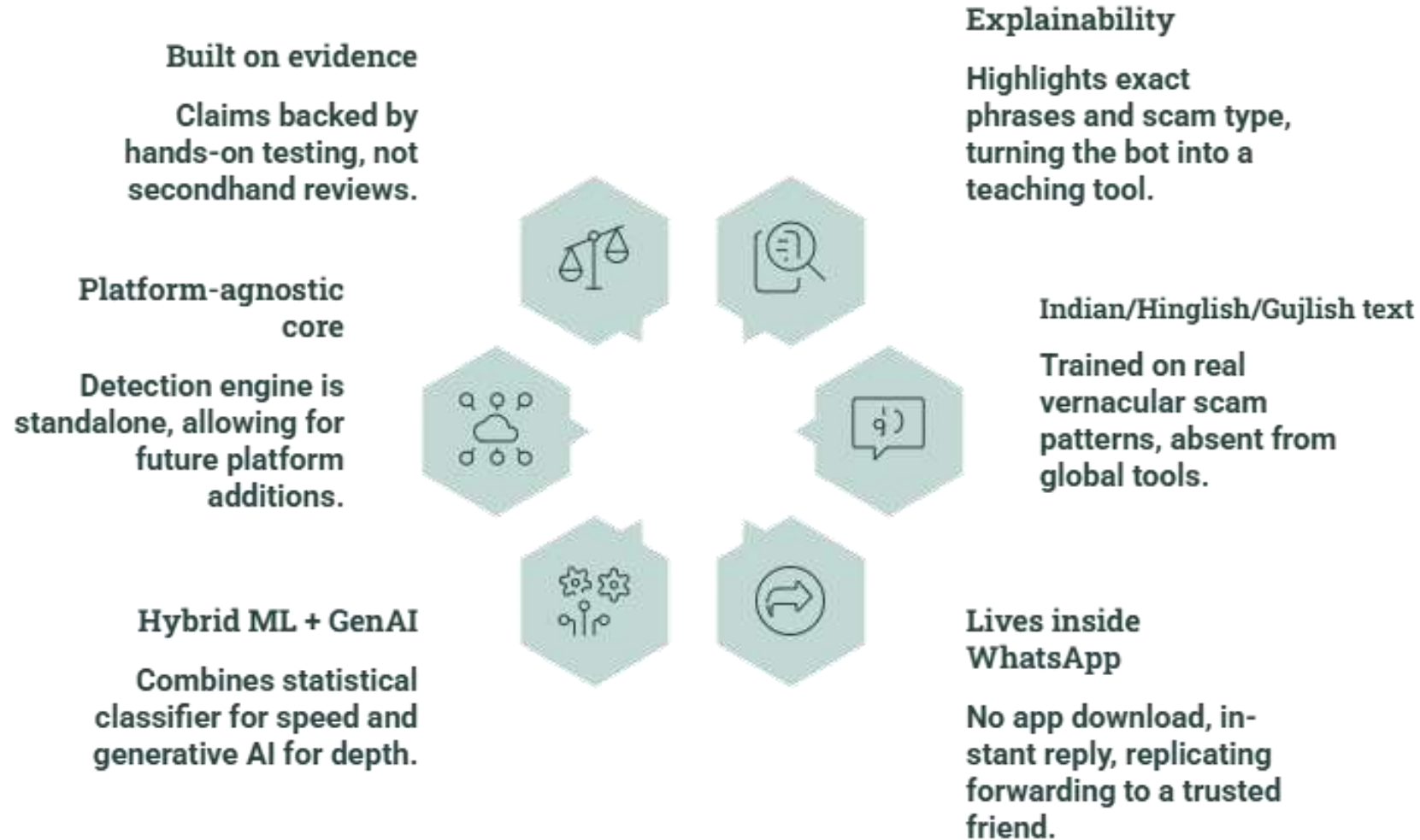
Generic, templated explanations: Even when correctly flagged, explanations never named the specific scam type (e.g., "digital arrest"), never quoted the actual phrases used, and never flagged the isolation tactic ("don't tell anyone").

Refuses to process plain conversational Hinglish: When sent a harmless personal message ("Bhai kal party mein aana bhool mat..."), Scamio did not analyze it — it flagged the text as being in "a different language" and asked the user to translate it into English before it could help.



Project Novelty

ScamShield AI

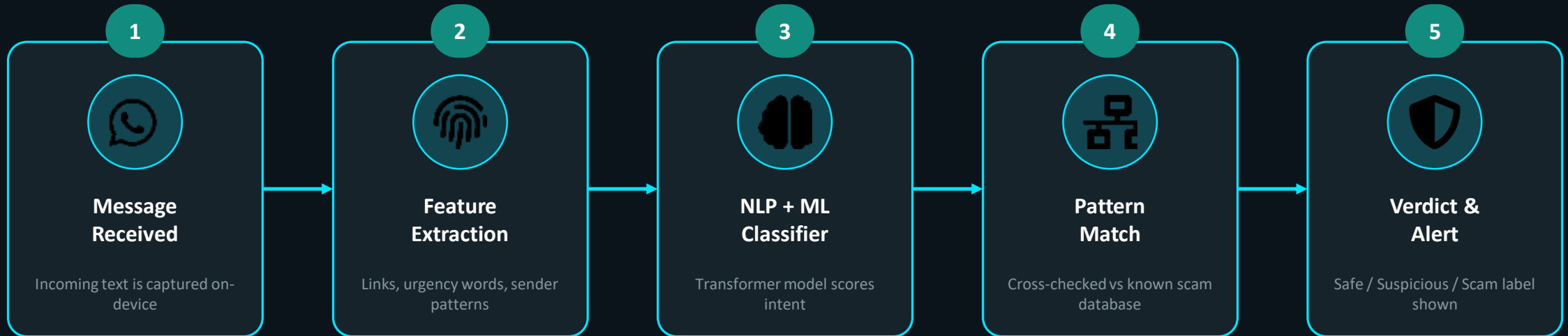


Technologies Used

Layer	Tool
Language	Python
Classic ML	scikit-learn (TF-IDF + Logistic Regression)
Generative AI	Anthropic / OpenAI API
API Framework	FastAPI
WhatsApp Connector	Twilio WhatsApp Sandbox API
Database	PostgreSQL
Dashboard	Streamlit
Hosting	Render (App + PostgreSQL)

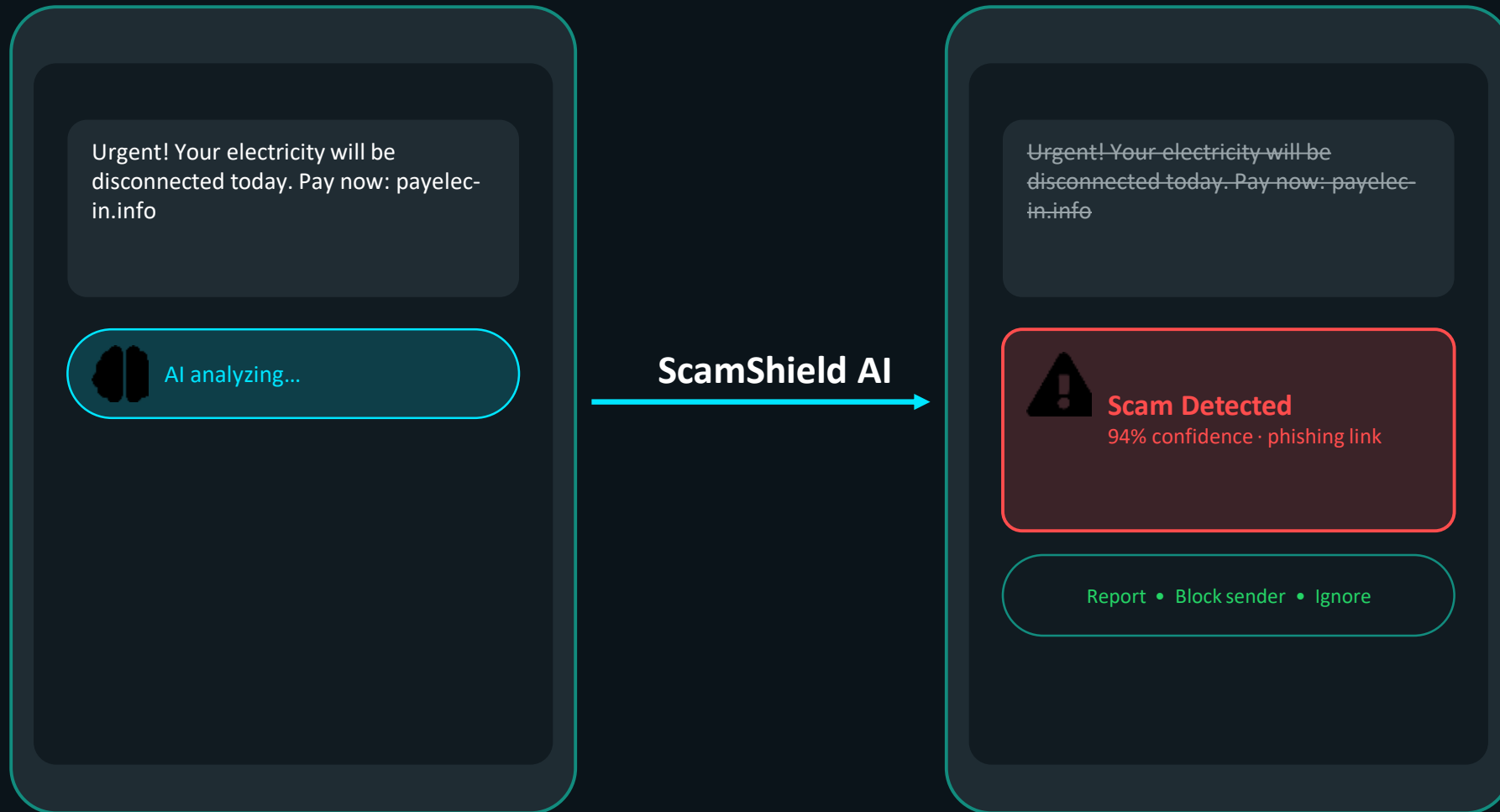
How the AI Detects a Scam

Every incoming message flows through a lightweight detection pipeline in real time.



See It In Action

A message arrives, gets scanned instantly, and the chat updates with a clear verdict.



Before — message just arrived

After — verdict delivered



Chat Smarter. Stay Safer.

ScamShield AI keeps the conversations you trust free of the messages you shouldn't.

Thank you, Any Questions?